Wenyu Chiou

Artificial Intelligence Evaluation & Behavioral Simulation Research Engineer

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Research engineer evaluating large language model (LLM) agents against measured human behavior and building quantitative behavioral simulations. Anchored by a **937-household** survey and a **52,141-household** coupled model.

Open to Summer 2027 internships in AI research, LLM evaluation, agent development, and computational social science.

Work authorization: F-1 student; CPT-eligible for internships.

EXPERIENCE

Graduate Research Assistant — Complex Adaptive Water Systems Lab, Lehigh University — Aug 2024 – present

- Designed a human-versus-LLM study using **937** household responses, multi-group structural equation modeling (SEM), and label-blind socioeconomic personas to compare generated behavior with measured pathways; planned submission to *Progress in Disaster Science*.
- Designed and fielded a **937-household** flood-adaptation survey; used SEM to identify social-group pathway differences and Bayesian calibration to translate observed pathways into agent parameters.
- Built **FLOODABM**, a **52,141-household** agent-based × catastrophe flood simulation across 27 census tracts (2011–2023) quantifying physical damage, insurance payouts, and household out-of-pocket exposure under NFIP mechanics.
- Designed and substantially completed the **Water Agent Governance Framework (WAGF)** to check LLM-agent decisions against physical, financial, and behavioral-theory constraints before state updates; in preparation for release.

Independent Research Engineering & Open Source — 2024 – present

- Built and maintain **ai-research-skills**, a 15-skill workflow for research gaps → literature synthesis → study design → paper drafting; built **codex-delegate**, whose regression test reproduces an upstream failure and verifies completion from git state.
- Shipped **research-hub**, a local-first workspace exposing Zotero, Obsidian, and NotebookLM through Model Context Protocol (MCP) and a command-line interface (CLI); published **research-hub-pipeline v1.1.1** on PyPI with cross-platform tests and fail-closed provenance checks.
- Created and self-hosted awesome-agentic-ai-zh, a trilingual agentic-AI roadmap with **240+ resources**, **23 exercises**, and **77 MCP/Skill entries**; reached **5K+ stars** and **700+ forks**.

SELECTED RESEARCH OUTPUTS

Yang, Y. C. E., & **Chiou, W.** (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111. DOI ***Published***

Chiou, W., et al. Household Flood Adaptation and Financial Outcomes: A Coupled Human–Flood Modeling Analysis of Homeowners and Renters. *Water Resources Research*. ***Under revision following peer review***

Chiou, W., et al. Flood Adaptation Decision Variables and Pathways across Social Groups: A Comparison between Human Survey Responses and Responses Generated by Large Language Models. Planned submission to *Progress in Disaster Science*.

SELECTED CONFERENCE PRESENTATIONS

International Society for Data Science and Analytics (ISDSA) 2026 Annual Meeting — “What Makes AI Believable?” Comparing AI and human responses in mental-health risk assessment; recorded presentation.

American Geophysical Union (AGU) Fall Meeting 2025 — Poster NH41E-0449: “Modeling Long-Term Household Flood Adaptation under Social Heterogeneity: A Coupled Agent-Based Modeling Framework.”

The 2nd International Sociohydrology Conference (ISHC) 2025, Tokyo — Oral Presentation 38-03: “Investigating the Decision-Making Process and Causal Relationships between Flood Risk Perception and Adaptation Actions.”

EDUCATION

Ph.D., Civil & Environmental Engineering, Lehigh University — 2024 – present · expected 2028

M.S., Hydrological and Oceanic Sciences, National Central University — Sep 2021 – Jun 2023

B.S., Earth Sciences, National Central University — Sep 2017 – Jun 2021

METHODS & TECHNICAL SKILLS

LLM / agent systems: LLM application & agent development · AI-system evaluation · evaluation infrastructure · prompt/context engineering · retrieval-augmented generation (RAG) · memory · tool use · multi-agent workflows · MCP

Methods: computational social science · behavioral simulation · experimental design · agent-based modeling (ABM) · structural equation modeling (SEM) · Bayesian calibration

Tools: Python · R · pytest / continuous integration (CI) · Git · OpenAI / Anthropic APIs · Ollama · Windows / POSIX